



TRAINING PROGRAMME

AWS CloudOps Engineer

Monitor, automate, deploy and keep AWS workloads secure and available

33 HOURS

FOR System administrators and operations engineers running workloads on AWS

PREPARES FOR AWS Certified CloudOps Engineer – Associate (SOA-C03)



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Overview

This programme teaches participants to operate workloads on Amazon Web Services and prepares the AWS Certified CloudOps Engineer – Associate exam (SOA-C03). They monitor and tune workloads with CloudWatch, X-Ray and Compute Optimizer, automate remediation with EventBridge and Systems Manager, keep services available with Auto Scaling, load balancing, AWS Backup and disaster recovery strategies, deploy with CloudFormation and the AWS developer tools, run containers on ECS and EKS, enforce security and cost controls, and troubleshoot networks. Almost every chapter ends with a lab, exam practice closes the content, and the programme ends with a final project: a workload operated end to end by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Monitoring & Automation	CloudWatch, X-Ray, performance tuning, EventBridge, Systems Manager	6 h
02	Reliability & Deployment	Auto Scaling, load balancing, AWS Backup, DR, CloudFormation, CI/CD, ECS and EKS	6 h
03	Security, Networking & Exam Preparation	IAM operations, Config, Security Hub CSPM, cost controls, VPC troubleshooting, SOA-C03	9 h
04	Final Project	A workload operated end to end	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
Total			33 h

Key facts

AUDIENCE	System administrators and operations engineers running workloads on AWS
PREREQUISITES	Core AWS services (course: AWS Cloud Practitioner) and the Linux command line (course: Linux & Bash Scripting), or equivalent. Core services are not taught again
DURATION	33 h: 27 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs; exam practice
LAB ENVIRONMENT	One laptop per participant with the AWS CLI; one AWS sandbox account per team, provided by the organiser, with a budget alert; stacks and recovery points deleted at the end of each lab to control cost
ASSESSMENT	A lab validated at the end of each phase, practice exam questions and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can deploy a workload with CloudFormation through a pipeline, monitor it, automate its remediation and patching, test its backup and recovery, report its cost and sit the SOA-C03 exam

PHASE 01

6 H

Monitoring & Automation

Participants make a workload observable with CloudWatch metrics, logs, Logs Insights, alarms, dashboards and X-Ray traces, tune it with Compute Optimizer and storage and database settings, then let AWS fix common problems on its own with EventBridge rules and Systems Manager runbooks. Validated lab: a disk-full alarm that triggers an Automation runbook which cleans the instance, with Session Manager replacing SSH.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Monitoring, Logging & Performance	CloudWatch metrics and the CloudWatch agent, logs, Logs Insights, alarms, dashboards, X-Ray; performance optimisation with Compute Optimizer, EBS and RDS tuning	Lecture + Lab	3 h
1.2	Remediation & Automation	EventBridge rules, Systems Manager Automation runbooks, Run Command, Patch Manager, Session Manager	Lecture + Lab	3 h

PHASE 02

6 H

Reliability & Deployment

Participants keep the workload available through failures and disasters with Auto Scaling, Elastic Load Balancing, multi-AZ designs, AWS Backup and recovery strategies chosen against RTO and RPO. They then deploy it as code with CloudFormation, StackSets and drift detection, through a pipeline built with the AWS developer tools, including container workloads on ECS and EKS. Validated lab: a CloudFormation stack deployed by CodePipeline, and a database restored from AWS Backup within the stated recovery objective.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Reliability & Business Continuity	Auto Scaling, Elastic Load Balancing, multi-AZ, AWS Backup, RTO and RPO, disaster recovery strategies	Lecture + Lab	3 h
2.2	Deployment, Provisioning & Automation	CloudFormation, StackSets, drift detection, CodeBuild, CodeDeploy, CodePipeline with a Git source (GitHub through AWS CodeConnections, or CodeCommit), container workloads on ECS and EKS	Lecture + Lab	3 h

PHASE 03

9 H

Security, Networking & Exam Preparation

Participants operate security and compliance controls with IAM, Organizations, Config rules, Security Hub CSPM and encryption, keep costs visible with Trusted Advisor, Cost Explorer and Budgets, troubleshoot connectivity with Flow Logs and Reachability Analyzer, then practise exam questions across the SOA-C03 domains. Validated lab: an unreachable instance diagnosed with Reachability Analyzer and fixed, and a non-compliant resource flagged by a Config rule.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Security & Compliance	IAM operations, Organizations, Config rules, Security Hub CSPM, encryption, Trusted Advisor, Cost Explorer and Budgets	Lecture + Lab	3 h
3.2	Networking & Content Delivery	VPC operations, Flow Logs, Reachability Analyzer, Route 53, CloudFront, connectivity troubleshooting	Lecture + Lab	3 h
3.3	SOA-C03 Preparation	Practice questions and review	Lecture + Workshop	3 h

PHASE 04

6 H

Final Project

Teams of two or three take charge of a provided workload, reusing the phase labs, then present how they run it as they would to an operations manager.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Deploy	Read the brief, deploy the workload with CloudFormation through CodePipeline, build dashboards and alarms	Project	3 h
4.2	Automate & Recover	Automated remediation and patching, a backup and disaster recovery test, the cost report and runbook	Project	1.5 h
4.3	Demo & Review	Live incident handled by automation, recovery results, teardown; questions and feedback	Project	1.5 h

FINAL PROJECT

Operating a Workload on AWS

A workload deployed with CloudFormation through CodePipeline, watched by dashboards and alarms, remediated and patched automatically, recovered in a disaster recovery test, and reported with its monthly cost.

STACK AWS: CloudFormation, CodePipeline, CodeBuild, CloudWatch, EventBridge, Systems Manager, AWS Backup, Auto Scaling, Elastic Load Balancing, Cost Explorer

DELIVERABLE Templates and pipeline, dashboards, runbook, DR test results, cost report, live demo